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TODO scripts improvement to address critics reviewers

1 message

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<https://chatgpt.com/share/69fa5df0-df40-8328-b1a3-336aa6ccfbad>

K+ Pumps Data-20260424T121023Z-3-001.zip

astro_atf_analysis_improved_
sectioned_stats_second_layer (1).ipynb

Filtered_baseline_sweep (1).ipynb

GPT_COMP_FIRST

Define “degeneracy” vs “non-identifiability” vs “sloppiness” with operational tests. (include modes>mechanisms mapping)Goal: Stop arguing degeneracy purely from “many parameter sets fit V_m .” Instead, show:

- * which parameters are non-identifiable from V_m alone;
- * whether the good-fit set is a manifold/continuum (compensation) vs separated modes;
- * whether different modes correspond to meaningfully different mechanisms (current/flux decomposition) while preserving function (K homeostasis).

Computations

1. Structural identifiability check
Use a tool like STRIKE-GOLDD-style analysis or symbolic/numerical methods to ask: in principle (ideal noise-free continuous V_m), are parameters identifiable? <https://doi.org/10.1371/journal.pcbi.1005153>
2. Practical identifiability via profile likelihood
Use a profile-likelihood workflow to compute confidence intervals and show which parameters (or combinations) are constrained by your data. <https://doi.org/10.1371/journal.pcbi.1011515>
3. Sloppiness (Fisher Information Matrix spectrum)
Compute FIM eigenvalues around the best fit(s) and show whether you have a “sloppy spectrum” (broad log-spaced eigenvalues) <https://doi.org/10.1371/journal.pcbi.1006694> <https://doi.org/10.1371/journal.pcbi.0030189> <https://doi.org/10.1371/journal.pcbi.1010844>
4. Map “modes” to mechanisms
For each cluster/mode, compute time courses of:
 - * Kir current contribution, gap junction flux contribution, leak contribution: Marder style smooth change of shade, compensation between two mode
 - * Net K clearance partitioning: Then show that clusters correspond to distinct mechanistic balances, not just numerically different parameters.

Manuscript deliverables

A short “Definitions” paragraph:

*non-identifiability = multiple parameter sets yield similar V_m because data cannot constrain them
 *sloppiness = broad sensitivity spectrum where predictions depend on stiff combinations
 *degeneracy (your claim) = multiple mechanistically distinct regimes that preserve K buffering function across perturbations

One main figure:

profile likelihood or FIM spectrum + mechanistic decomposition by cluster.

Parameter plausibility and interpretation (“effective parameters” vs physiological) +

mechanistic interpretability**Goal**

Address the criticism that some fitted parameters may be out of physiological range and that fitting V_m alone weakly constrains ionic dynamics.

Computations

Report fitted parameter distributions per condition and per cluster:
medians, IQRs, and full ranges: "within biophysical range,"

Constrained inference

Re-run optimization with tighter physiological priors/ranges (or penalty terms).
Show whether fit quality and degeneracy claims persist.

Mechanistic interpretability

d , can be considered as an effective param; $gt*S_a$ and $gs*S_a$, also can be as effective params,
If a parameter consistently goes "out of range," explicitly label it as an effective parameter arising from model reduction (and quantify how).

Parameter plausibility: quantify fitted parameter distributions vs physiological ranges; if some are "effective parameters," show that explicitly and/or constrain with priors/ranges.

Robustness / predictive checks beyond fitted traces (including perturbation)**Goal:**

Show that your "degenerate" parameter sets are not only fitting artifacts but yield robust predictions.

Computations

1. Cross-validation across pump currents

Fit using 4–5 currents, predict the held-out current trace; rotate across all currents. This uses no new experiments.

2. Posterior predictive checks / prediction intervals

Even without full Bayesian inference, you can propagate uncertainty from accepted parameter sets to prediction bands (simulate ensemble). PLOS Comp Biol parameter inference papers emphasize diagnostic checks and assessing uniqueness/quality of inferred distributions. <https://doi.org/10.1371/journal.pcbi.1011108>

3. Perturbation tests (in silico):

vary bath coupling ϵ vary stimulus duration vary baseline K_o
Compare whether different parameter clusters diverge in predictions—this is where "distinct phenotypes/mechanisms" becomes testable computationally.

Manuscript deliverables

A "Predictive validation" supplement: cross-validated errors and prediction bands by condition.

A reframed conclusion: robust predictions (or explicit limitations if robustness fails).

Stress-test key modeling assumptions computationally (without adding experiments)

Goal: -manuscript explicitly motivates: (i) sigmoid gating of spatial buffering; (ii) intracellular K accumulation as proxy for ECS K; (iii) splitting intracellular K into "local" and "syncytial" components. *Show your conclusions (especially degeneracy claims) are not brittle to these modeling choices.

Computations

1. Alternative gating functions (model comparison)

*Keep the same number of parameters but change form: sigmoid vs Hill vs soft-threshold linear.

*Refit and compare fit metrics (RMSE/AIC) and whether clustering/multimodality persists. <https://doi.org/10.1371/journal.pcbi.1000696>

2. Proxy validation

Quantify the correlation between $\Delta K_{a,t}$ and ΔK_o across your simulations; show regimes where proxy fails.

If needed: add an explicit ECS compartmental relation (minimal extension) and show results are unchanged qualitatively.

3. Compartment-split sensitivity

Try an equivalent formulation with a single intracellular K state and compute whether inferred degeneracy structure changes.

Manuscript deliverables

Supplementary "assumption sensitivity" section with 2–3 key comparisons.

In Discussion: clear boundary of what the minimal model can/can't support.

Figure/organization issues that are computational (replotting, consistent axes/units, reducing clutter, clearer layouts).

EXPERIMENTAL DATA regarding phenotypes

3 pièces jointes



SuggestionsAdressingCritics.md
21K



AFT_uncut_end_BARIUM_Opt_v2 (1).ipynb
3323K



astro_atf_analysis_improved_sectioned.ipynb
13326K